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BASIC MATHEMATICS

FORM TWO

Form Two National Assessment (FTNA) 2023 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Two
Year	2023
Exam Board	NECTA
Exam	Form Two National Assessment (FTNA)
Type	Answer key and worked solutions
Questions	10 compulsory questions; 23 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F2-2023.pdf

Question 1 — Multiples and approximation

1(a). List the first twelve multiples of 4 and 5; identify the common multiples.

Step 1: Multiply 4 by each integer 1-12; repeat with 5.

Step 2: Compare both lists and keep the values appearing in each. The LCM is 20; 60 is not in the first twelve multiples of 4.

Final Answer: 4: 4,8,12,16,20,24,28,32,36,40,44,48. 5: 5,10,15,20,25,30,35,40,45,50,55,60. Common in the lists: 20 and 40.

1(b)(i). Evaluate $(2/25) \times 0.737$ to one significant figure.

Step 1: $2/25 = 0.08$; exact product = $0.08 \times 0.737 = 0.05896$.

Step 2: The first significant digit is 5 and the next digit is 8; round it up to 6 in the hundredths place.

Final Answer: 0.06.

1(b)(ii). Give the same product to three decimal places.

Step 1: The exact product is 0.05896.

Step 2: Keep three digits after the decimal point, 058; the fourth digit is 9, so round 058 up to 059.

Final Answer: 0.059.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 2 — Fractions and percentage change

2(a). Arrange $2/3$, $4/7$, $3/8$ and $5/9$ in ascending order.

Step 1: Use common denominator 504: $2/3 = 336/504$, $4/7 = 288/504$, $3/8 = 189/504$, $5/9 = 280/504$.

Step 2: Order numerators: $189 < 280 < 288 < 336$, then write the corresponding original fractions.

Final Answer: $3/8 < 5/9 < 4/7 < 2/3$.

2(b). Mericho's population was 2,800 in 2016 and increased by 8% in 2017. Find the 2017 population.

Step 1: Increase = $(8/100) \times 2800 = 224$.

Step 2: New population = $2800 + 224 = 3024$.

Final Answer: 3024 people.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 3 — Units, profit and loss

3(a). 1000 tonnes of maize are shared equally among 25 schools. How many kilograms does each receive?

Step 1: 1000 tonnes = 1,000,000 kg because 1 tonne = 1000 kg.

Step 2: Divide 1,000,000 by 25 to obtain 40,000 kg.

Final Answer: 40,000 kg per school.

3(b). A radio cost Tsh 80,000 and was sold at 20% profit. Find profit and selling price.

Step 1: Profit percentage is calculated on cost: $0.20 \times 80,000 = 16,000$.

Step 2: Selling price = cost + profit = $80,000 + 16,000 = 96,000$.

Final Answer: Profit: Tsh 16,000; selling price: Tsh 96,000.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 4 — Geometry

4(a). In trapezium ABCD, $AD \parallel BC$, $A = 60^\circ$, $D = 40^\circ$. At C, a is the upper exterior angle and c the interior angle; $B = b$. Find a, b, c .

Step 1: Co-interior angles on AB give $b + 60 = 180$, hence $b = 120^\circ$.

Step 2: Co-interior angles on CD give $c + 40 = 180$, hence $c = 140^\circ$.

Step 3: a and c form a straight angle, so $a = 180 - 140 = 40^\circ$ (also alternate interior to D).

Final Answer: $a = 40^\circ$, $b = 120^\circ$, $c = 140^\circ$.

4(b). A square room has side 5 m. Find its perimeter and area.

Step 1: Perimeter = $4 \times \text{side} = 4 \times 5 = 20$ m.

Step 2: Area = $\text{side}^2 = 5^2 = 25$ m². Use squared units for area.

Final Answer: Perimeter = 20 m; area = 25 m².

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 5 — Algebra

5(a). If $(6y + 1)/4 = 5(y + 5)/6$, find y to three significant figures.

Step 1: Multiply by 12 to remove denominators: $3(6y + 1) = 10(y + 5)$.

Step 2: Expand: $18y + 3 = 10y + 50$, so $8y = 47$ and $y = 47/8 = 5.875$.

Step 3: Retain the first three significant digits 5,8,7; the following 5 rounds 7 up to 8.

Final Answer: $y = 5.88$ (exactly $47/8$).

5(b). Solve $3x^2 - 7x - 6 = 0$ by completing the square.**Step 1:** Divide by 3: $x^2 - (7/3)x = 2$.**Step 2:** Add $(7/6)^2$ to both sides: $(x - 7/6)^2 = 2 + 49/36 = 121/36$.**Step 3:** Take both square roots: $x - 7/6 = \pm 11/6$, so $x = 18/6 = 3$ or $-4/6 = -2/3$.**Final Answer: $x = 3$ or $x = -2/3$.**

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 6 — Coordinates and reflection**6(a). Find the gradient through $(-1, 2)$ and $(3, -5)$.****Step 1:** Gradient = change in y divided by change in x.**Step 2:** $m = (-5 - 2)/(3 - (-1)) = -7/4$.**Final Answer: $-7/4$.****6(b). Find the image of $P(-3, 7)$ after reflection in the x-axis and y-axis.****Step 1:** Reflection in the x-axis maps (x, y) to $(x, -y)$; reflection in the y-axis maps (x, y) to $(-x, y)$.**Step 2:** Apply each to P for separate images. If both transformations are applied one after the other, both signs change.**Final Answer: In the x-axis: $(-3, -7)$. In the y-axis: $(3, 7)$. If applied successively in both axes: $(3, -7)$.**

Study Tip: The wording can mean separate reflections or successive reflections, so the distinction is shown explicitly.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 7 — Indices and logarithms

7(a). Solve $16^{(3 - n)} \times 2^{(1 + n)} = 1/2$; express n as an improper fraction.

Step 1: Rewrite $16 = 2^4$ and $1/2 = 2^{-1}$.

Step 2: Combine equal bases: $2^{[4(3 - n) + (1 + n)]} = 2^{-1}$.

Step 3: Equate exponents: $13 - 3n = -1$, giving $3n = 14$ and $n = 14/3$. Check $4(3 - 14/3) + 1 + 14/3 = -1$.

Final Answer: $n = 14/3$.

Study Tip: The faint first exponent is $3 - n$. The original wording and a clearer reproduction in the NECTA report confirm it.

7(b). Solve $\log(2x^2 + 1) + \log 4 = \log(7x^2 + 8)$.

Step 1: Combine logarithms: $\log[4(2x^2 + 1)] = \log(7x^2 + 8)$.

Step 2: Equal logarithms of positive quantities have equal arguments: $8x^2 + 4 = 7x^2 + 8$.

Step 3: $x^2 = 4$, so $x = \pm 2$. Both arguments are positive for these values, so both are valid.

Final Answer: $x = 2$ or $x = -2$.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 8 — Similarity and congruence

8(a). $DE \parallel BC$, $AD:BD = 3:5$ and $AC = 9.6$ cm. Find AE .

Step 1: AB contains $3 + 5 = 8$ ratio parts, so $AD/AB = 3/8$.

Step 2: $\triangle ADE \sim \triangle ABC$ by AA; $AE/AC = AD/AB = 3/8$.

Step 3: $AE = (3/8) \times 9.6 = 3.6$ cm.

Final Answer: $AE = 3.6$ cm.

Study Tip: Use $AD/AB = 3/8$, not $AD/BD = 3/5$, when comparing the small triangle to the whole triangle.

8(b). $AB = AC$ and D is the midpoint of BC . Prove $\angle ABD = \angle ACD$.

Step 1: $AB = AC$ is given; $BD = DC$ because D is the midpoint; AD is shared.

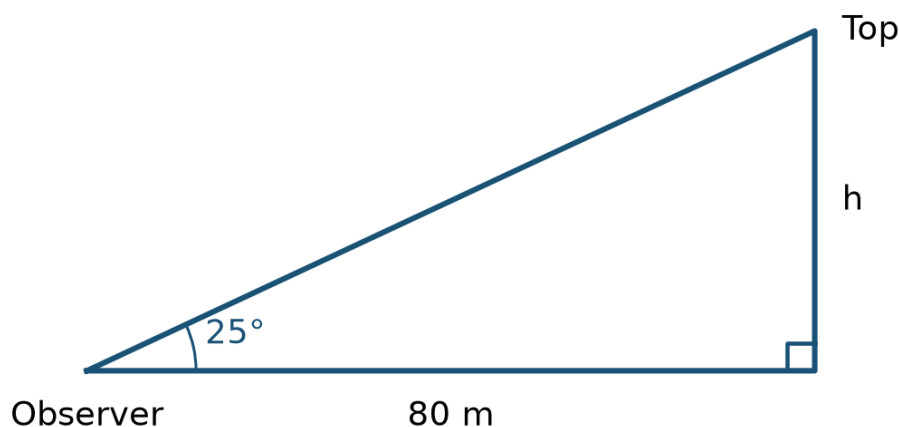
Step 2: Therefore $\triangle ABD \cong \triangle ACD$ by SSS. Corresponding angles at B and C are equal, establishing the required angle equality.

Final Answer: $\angle ABD = \angle ACD$.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 9 — Trigonometry and Pythagoras

9(a). A point is 80 m from a building and its angle of elevation is 25° . Find the height to one decimal place.



Step 1: Use the right triangle: height is opposite 25° and ground distance 80 m is adjacent.

Step 2: $\tan 25^\circ = h/80$; $h = 80 \tan 25^\circ \approx 37.3046$ m.

Step 3: To one decimal place, the height is 37.3 m.

Final Answer: 37.3 m.

9(b). In right triangle ABC, AB = 6 m, AC = 8 m and the right angle is at A. Find BC.

Step 1: BC is the hypotenuse; apply Pythagoras: $BC^2 = AB^2 + AC^2$.

Step 2: $BC^2 = 6^2 + 8^2 = 100$; take the positive square root to obtain 10 m.

Final Answer: 10 m.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Question 10 — Sets and statistics

10(a). 250 students attended the first day, 350 the second, 150 both, and 10 neither. Without a Venn diagram find total enrolment.

Step 1: Add the daily totals, then remove the overlap counted twice: attended at least once = $250 + 350 - 150 = 450$.

Step 2: Include students absent on both days: total = $450 + 10 = 460$.

Final Answer: 460 students.

10(b)(i). The pie-chart sectors are Kiswahili 210° , Mathematics 120° , Chemistry 30° . Find the fraction who passed Kiswahili.

Step 1: A complete pie chart represents 360° .

Step 2: Fraction = $210/360 = 7/12$.

Final Answer: $7/12$.

10(b)(ii). Find the percentage who passed Mathematics.

Step 1: Mathematics fraction = $120/360 = 1/3$.

Step 2: Multiply by 100%: $(1/3) \times 100\% = 33\frac{1}{3}\%$.

Final Answer: $33\frac{1}{3}\%$.

10(b)(iii). Find the percentage who passed Mathematics and Chemistry, as represented by their combined sectors.

Step 1: Add the sector angles: $120 + 30 = 150^\circ$.

Step 2: Percentage = $(150/360) \times 100\% = (5/12) \times 100\% = 41\frac{2}{3}\%$.

Final Answer: $41\frac{2}{3}\%$.

Study Tip: The pie chart partitions students into separate categories. Here “and” asks for the combined categories; it does not provide data on individuals passing both subjects.

Section Summary: Use the stated method, retain exact values until the final rounding, and check the original diagram or equation before calculating.

Source Checks

NECTA — FTNA 2023 Basic Mathematics: original index-equation notation